

Multimodal Data of the 2026 Portuguese Presidential Elections

Exploratory Data Analysis

Group 25:

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Processing Big Data

Data Overview



Visual

- Bounding boxes + confidence
- Poses
- Facial emotions (8 Classes)



Audio/Speech

- Acoustic Features (Pitch, jitter)
- Speaker diarization
- Text transcripts & embedding



Original source

- 28 political debates
- 7 newscasts
- Multi source polling data

Visual data noise

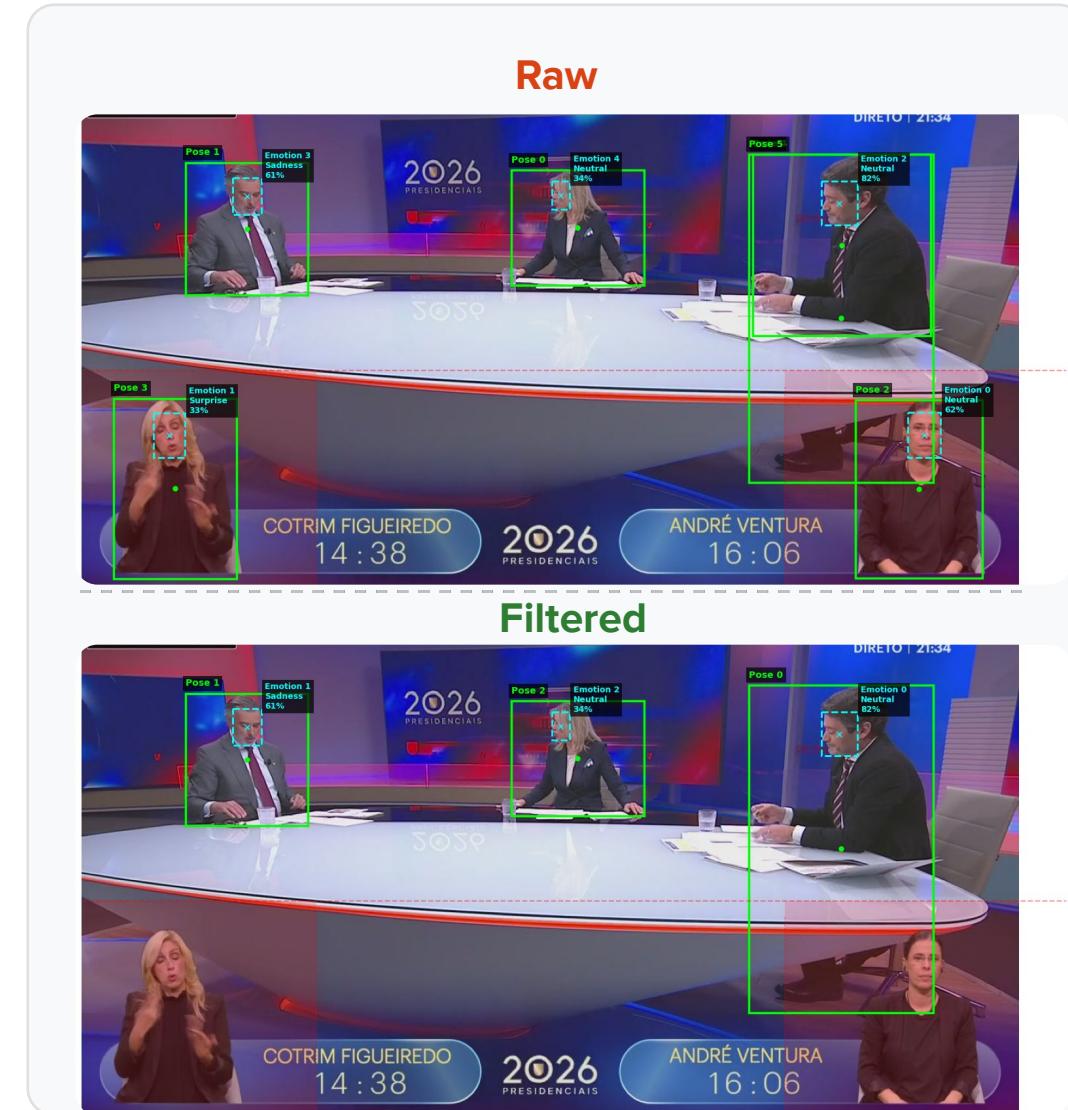
- Unsorted entries → Sorted entries by n° of frame

Noise Sources

- Sign language interpreters
- Bounding box overlapping duplicates

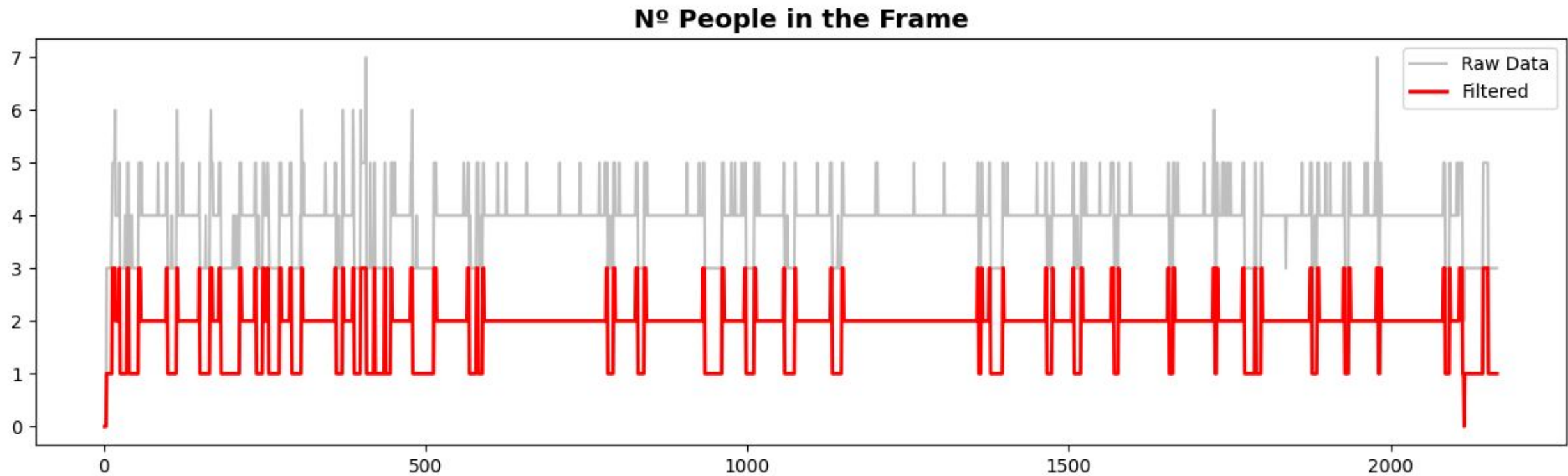
Solution

- Excluding corner dead zones
- Confidence threshold > 40%
- Exclusion by overlapping
- Exclusion by area size of bounding box



Visual data noise

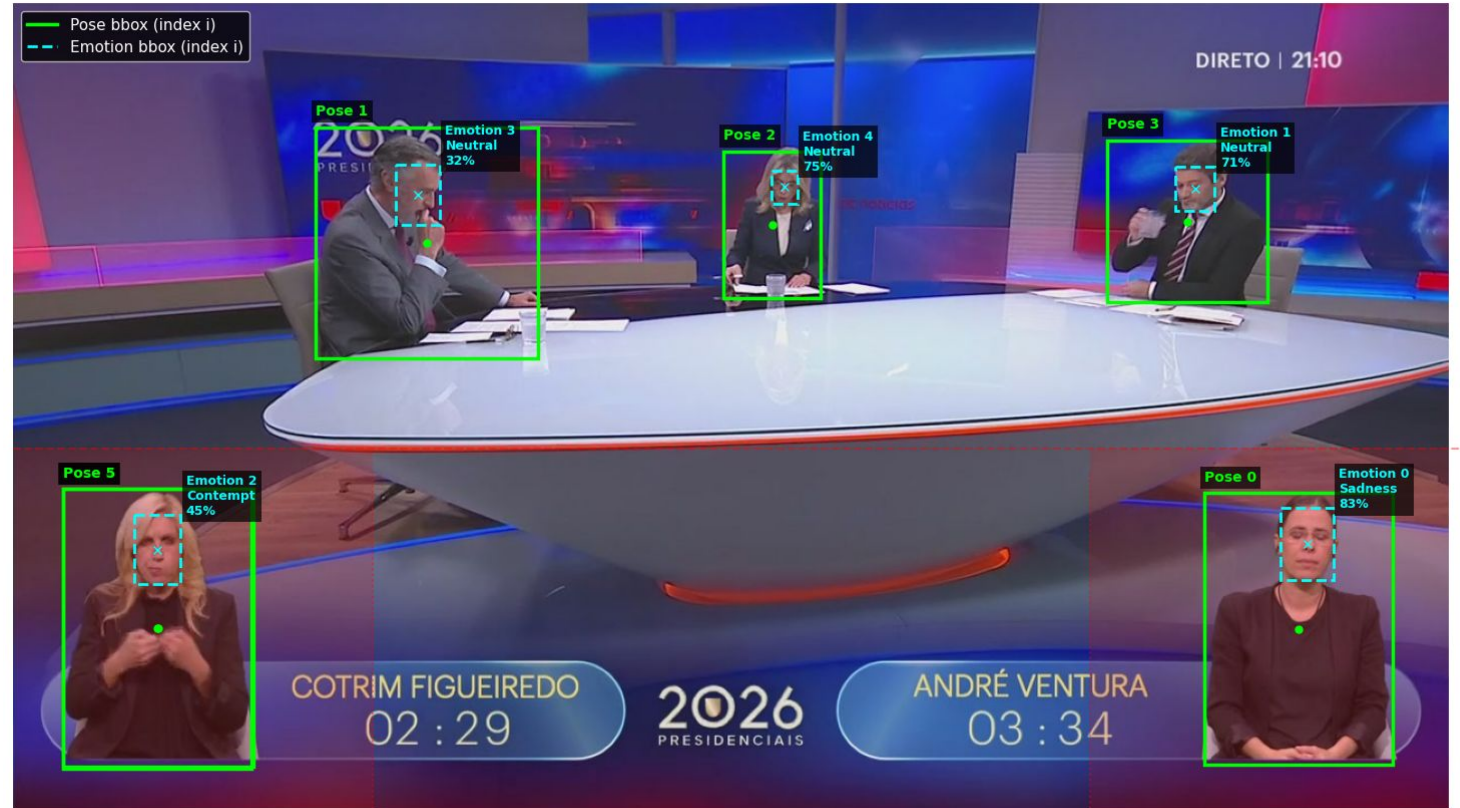
Example for Cotrim Figueiredo vs Ventura debate on December 19



- NºPeople = length of cleaned poses vector

Missing data is information

0 faces $\approx$ camera cuts, wide shots or commercials — not model failure
Frames and audio are sampled **independently**
→ timelines not aligned
Some sections lack a transcript or an emotion (**model gaps**)
Treated as a **domain feature**, handled when merging modalities



Pose + emotion detections overlaid on a debate frame

2D Visualization of Audio Characteristics

Final Goal: How to identify a segment transcript speaker using the audio

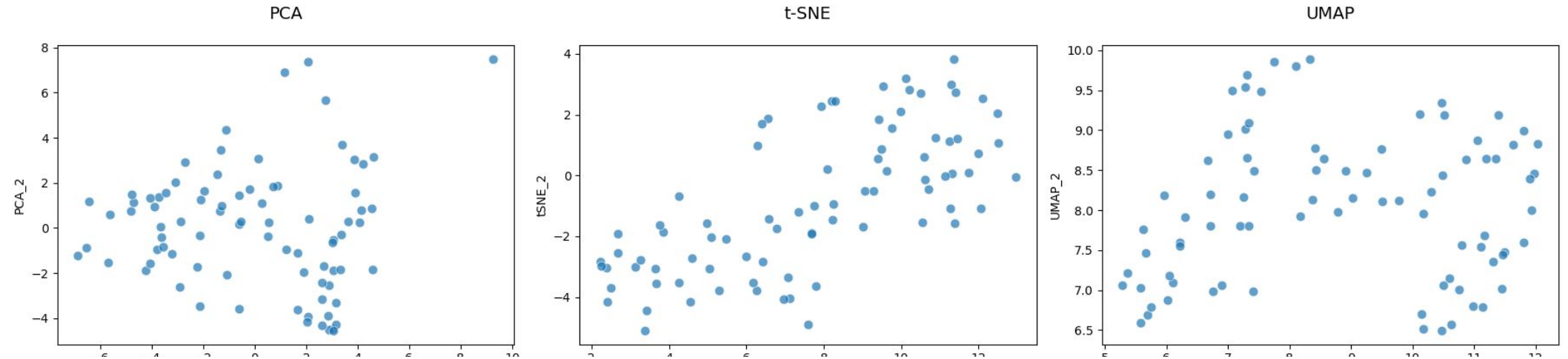
- Which audio characteristics are more ideal to make this?
 - Simple acoustic features: pitch, speak rate,...
 - Pyannote speaker embeddings

Use dimensionality reduction to 2D to look for possible Clusters

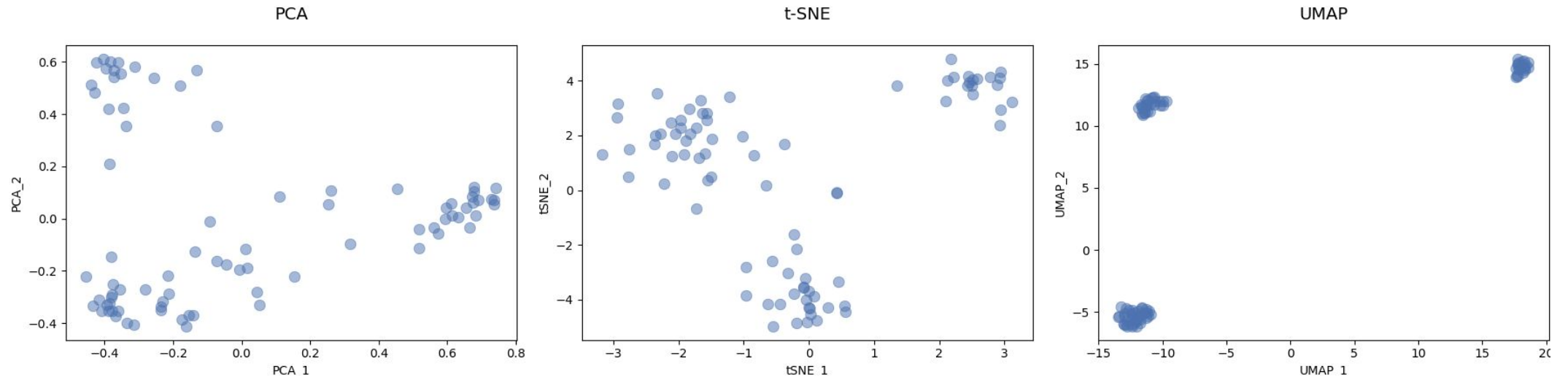
- Which algorithm to use?
 - Simple PCA
 - T-SNE
 - UMAP

2D Visualization of Audio Characteristics

Dimensionality Reduction: Only Acoustic Features (30D -> 2D)



Dimensionality Reduction: Speaker Embeddings (512D -> 2D)



Single Video Analysis

Selected Datasets



Debate

Seguro vs Filipe

December 19



Newscast

Telejornal RTP

December 2

Moment Segmentation

Extraction Rule

- **Count = 1:** Checkbox center (Left vs. Right)
- **Count = 2:** Two-shot (Split screen or shared)
- **Count ≥ 3 :** Wide shot (Includes noise/interpreter)

State Classification

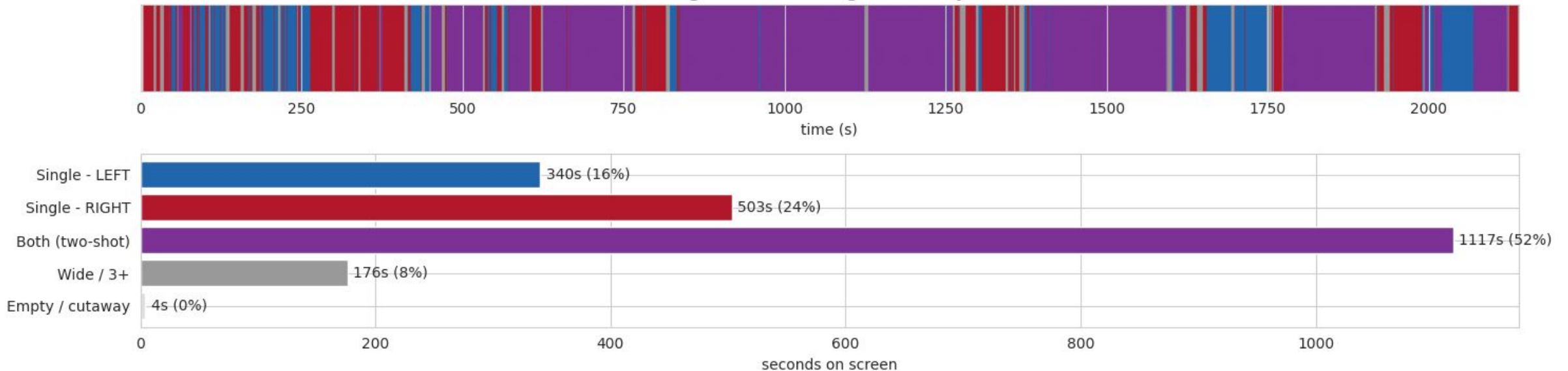
Single-LEFT: Person 1 solo close-up

Single-RIGHT: Person 2 solo close-up

Both: Shared screen time

Empty: Transitions / Camera cuts

Camera framing over time - Seguro vs Filipe December 19

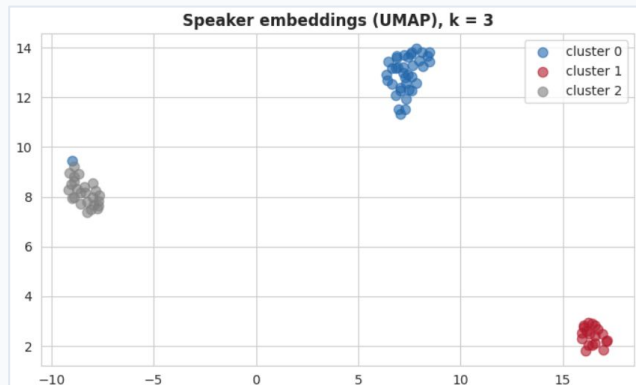


Cross-Modal Speaker Discovery

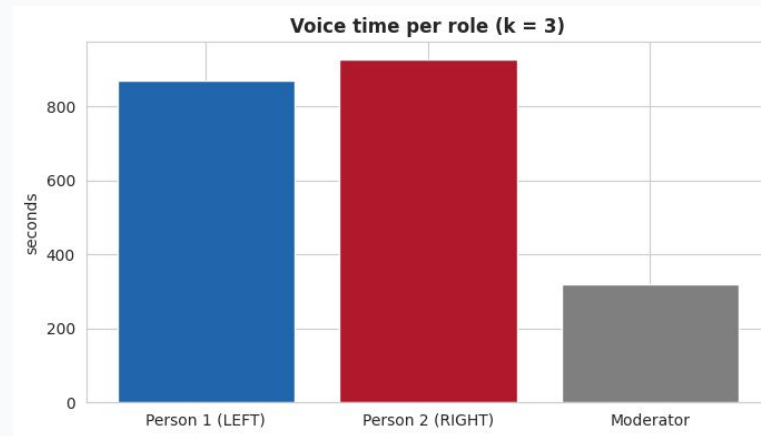
Zero-Manual-Label Validation

Audio knows nothing about the video, yet K-Means on the voiceprints recovers the same two people the pose detector sees on the left and right — and the third, quiet, side-neutral cluster is the moderator.

Voiceprint UMAP Cluster (k=3)



Total Speech Time per Voice Cluster



Clustering Strength (K=3)

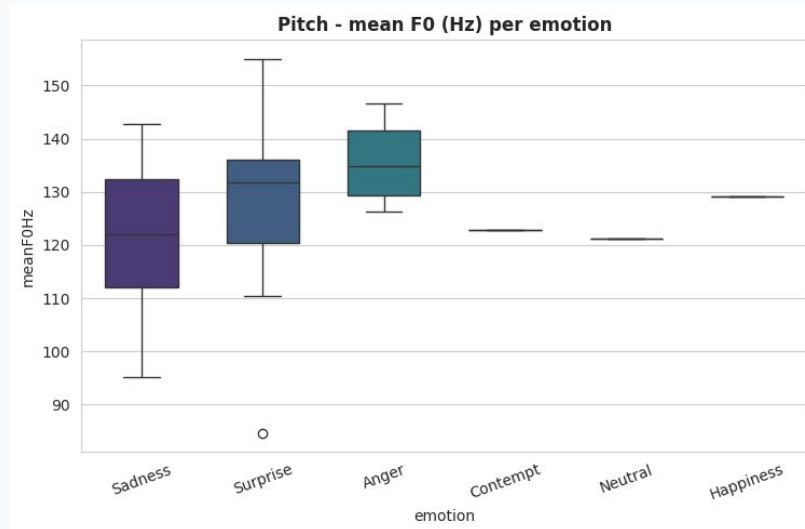


Audio ↔ Emotion Correlation

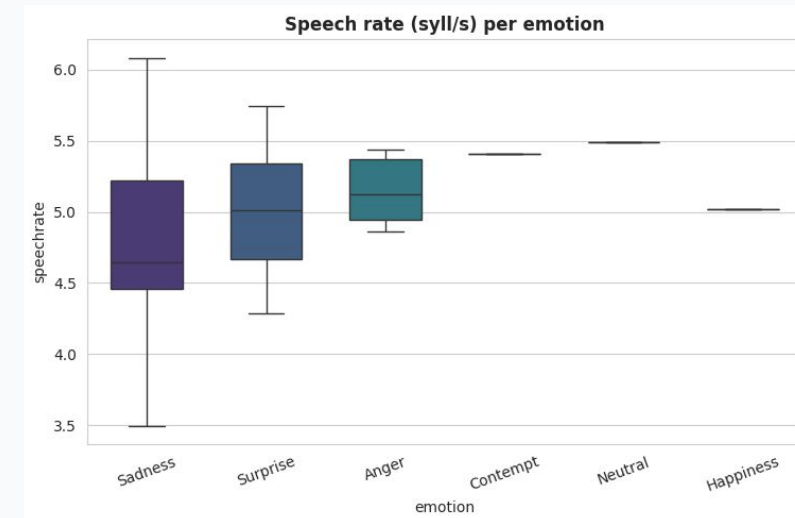
Cross-Modal Feature Pairing

- Read the speaking person's facial emotion → pair with pitch/rate

Pitch (Hz) Distribution per Emotion



Speech Rate (syll/s) per Emotion

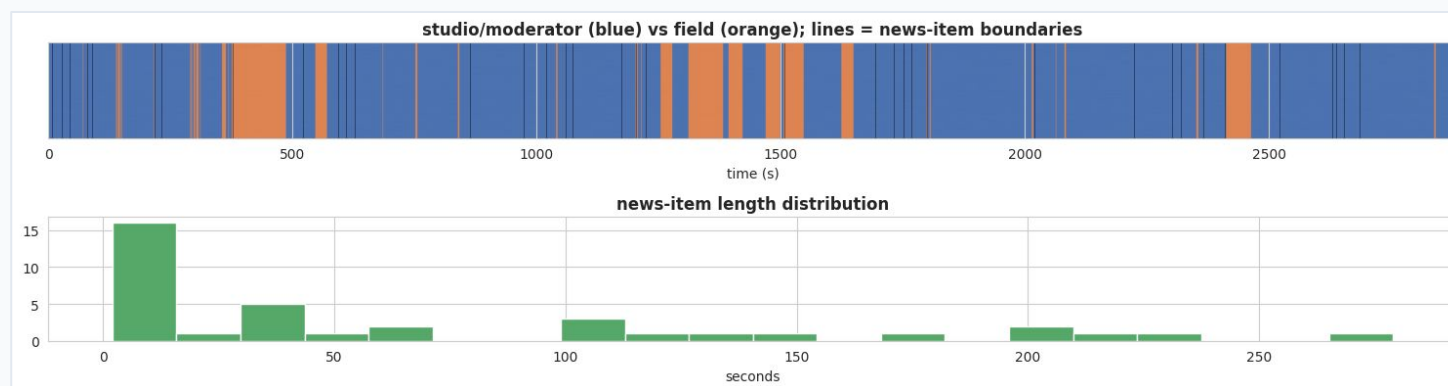


Newscast: Anchor ↔ News & Topics

Story Segmentation & OCR Labeling

- Banner (OCR) names each story; location/reporter tags mark field footage.
- Merge caption-runs → **≈37 news items**, median length **≈31 s**.

Multimodal Segmentation: Studio (Blue) vs. Field (Orange)



Segment Count

37

Total Stories

Median Duration

31s

per Topic

Cross-debate voice model detection

Automated Candidate Identification

- Each candidate has 7 debates
 - their voice reoccurs
 - opponents/moderator don't
- Iterative matching picks each candidate's consistent cluster across those debates
 - names the voice
 - isolates the moderator

Validation Accuracy

28/28

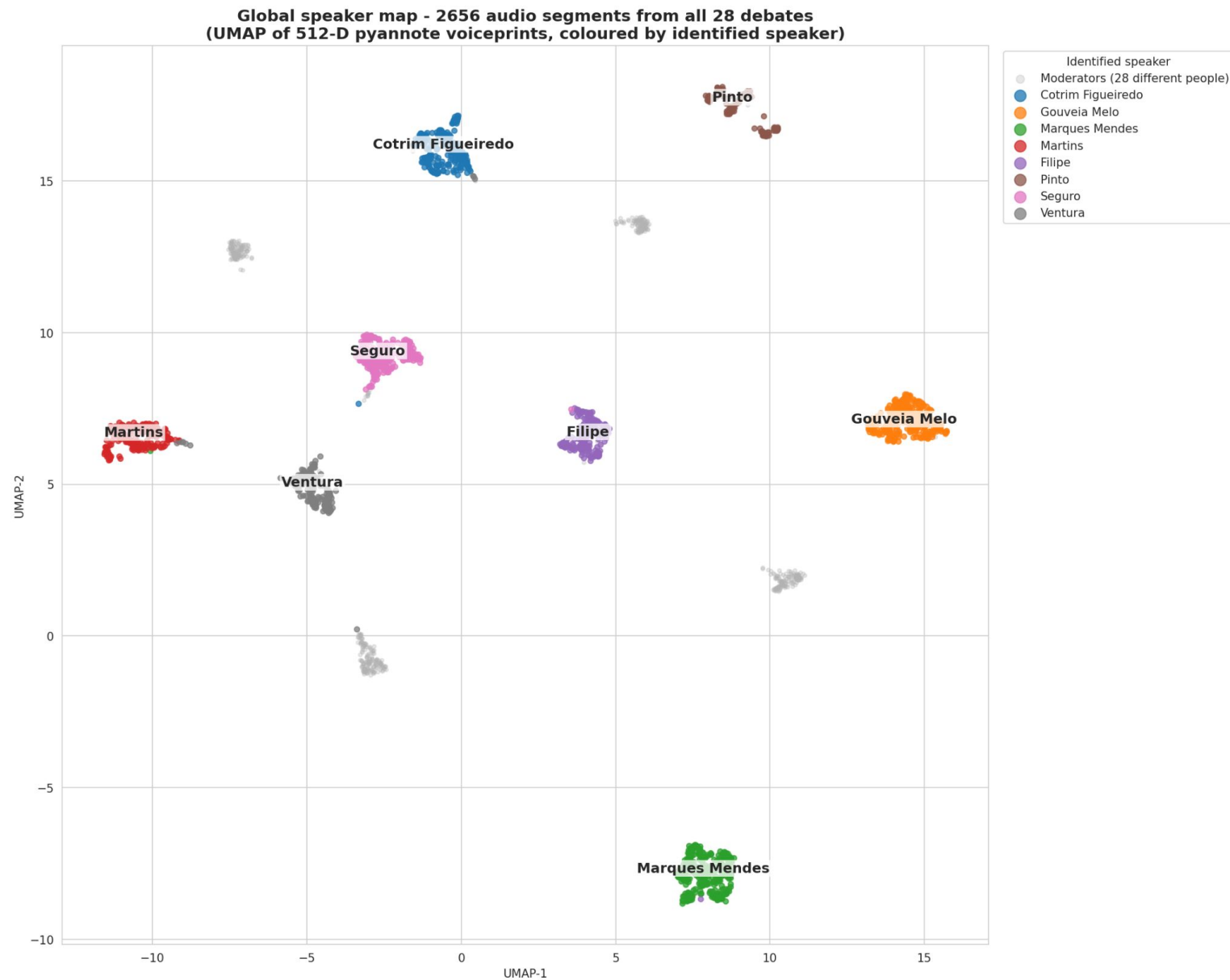
Distinct Clusters / Mod. Quietest

Voiceprint Consistency

0.89-0.91

Clustering Confidence Score

Cross-debate voice model detection



Vocabulary & vocal fingerprint

Distinctive Vocabulary Across Candidates

Ventura (CH)

- "Corruptos"
- "Castro"
- "Chacha"

Cotrim (IL)

- "Reuniões"
- "Migrações"
- "Exigência"

Filipe (PCP)

- "Solidário"
- "Palestina*"
- "Greve*"

Marques Mendes (AD)

- "Corrupção"
- "Prioridades"
- "Acusações"

Martins (BE)

- "Abandonado"
- "Zelenski"
- "Milionários"

Pinto (L)

- "Esquerdas"
- "Drástica"
- "Surgiu"

Seguro (PS)

- "Aplicação"
- "Planos"
- "Legislação"

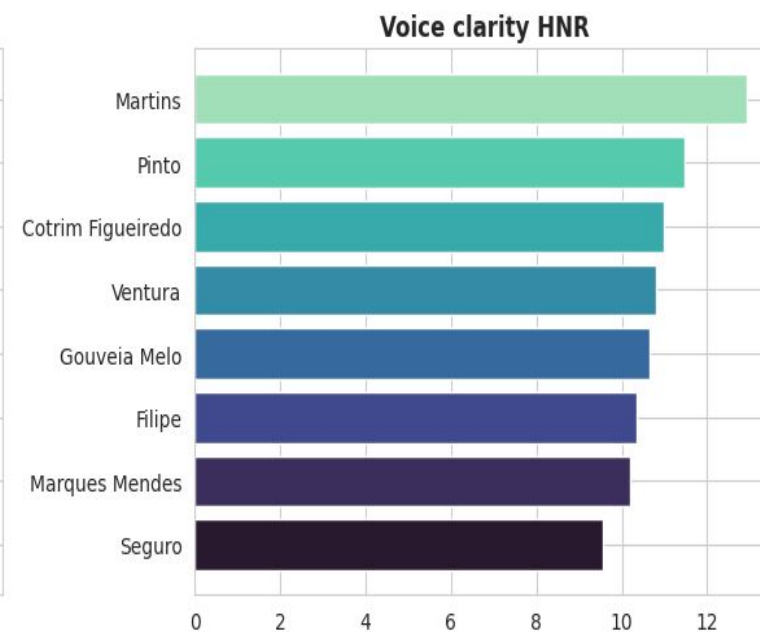
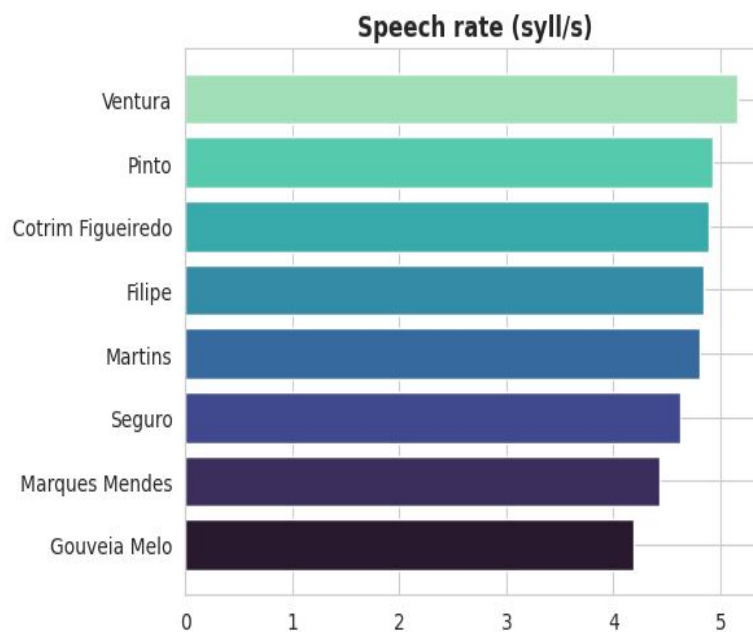
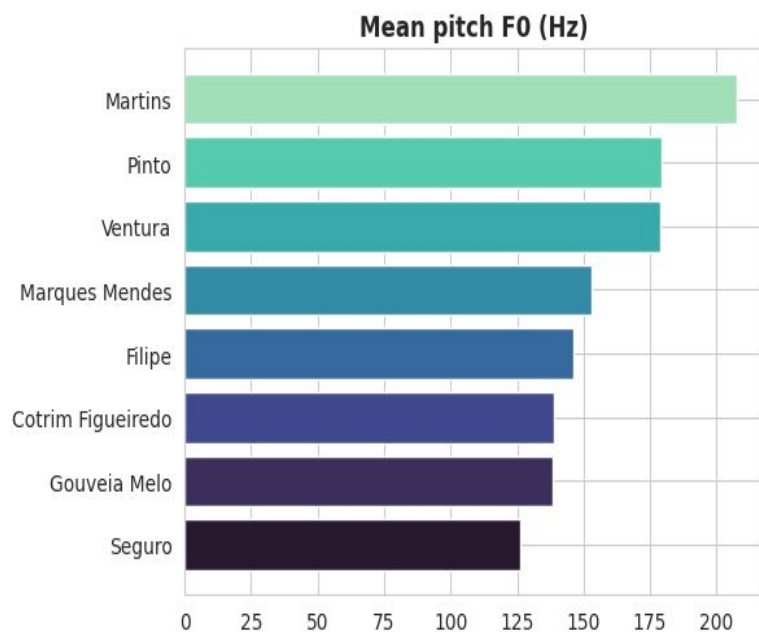
Gouveia Melo (-)

- "Marcos"
- "Governação"
- "Angola"

Vocabulary & vocal fingerprint

Vocal Profiles

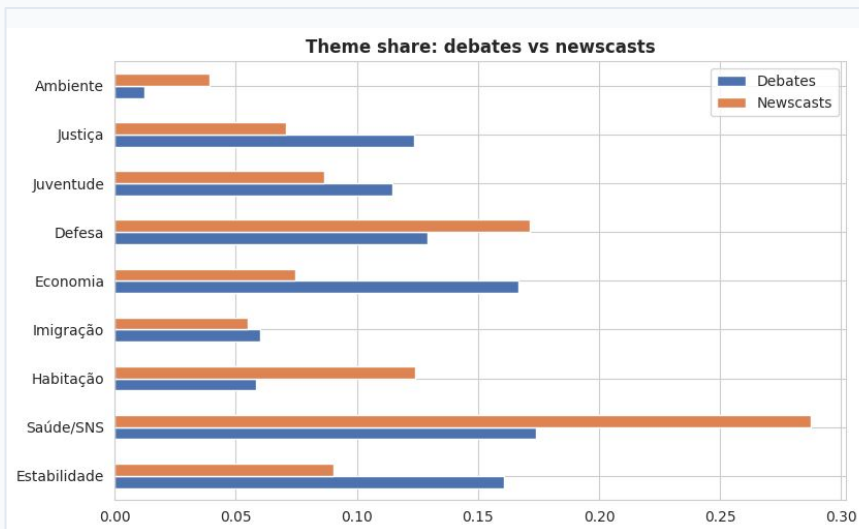
Per-candidate vocal profile (moderator excluded)



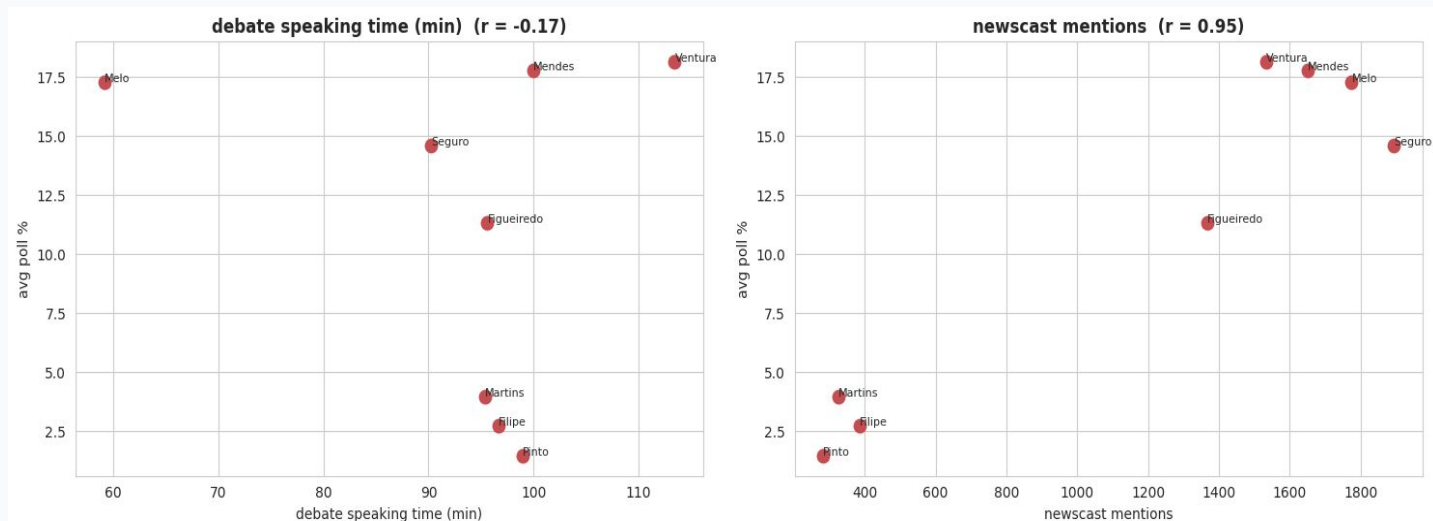
Press mirrors the election

- **Theme Overlap:** Debate vs. newscast theme no clear overlap on the themes.
- **Correlation:**
 - Newscast mentions vs. polls: $r \approx 0.95$
 - Debate speaking time vs. polls: $r \approx 0$

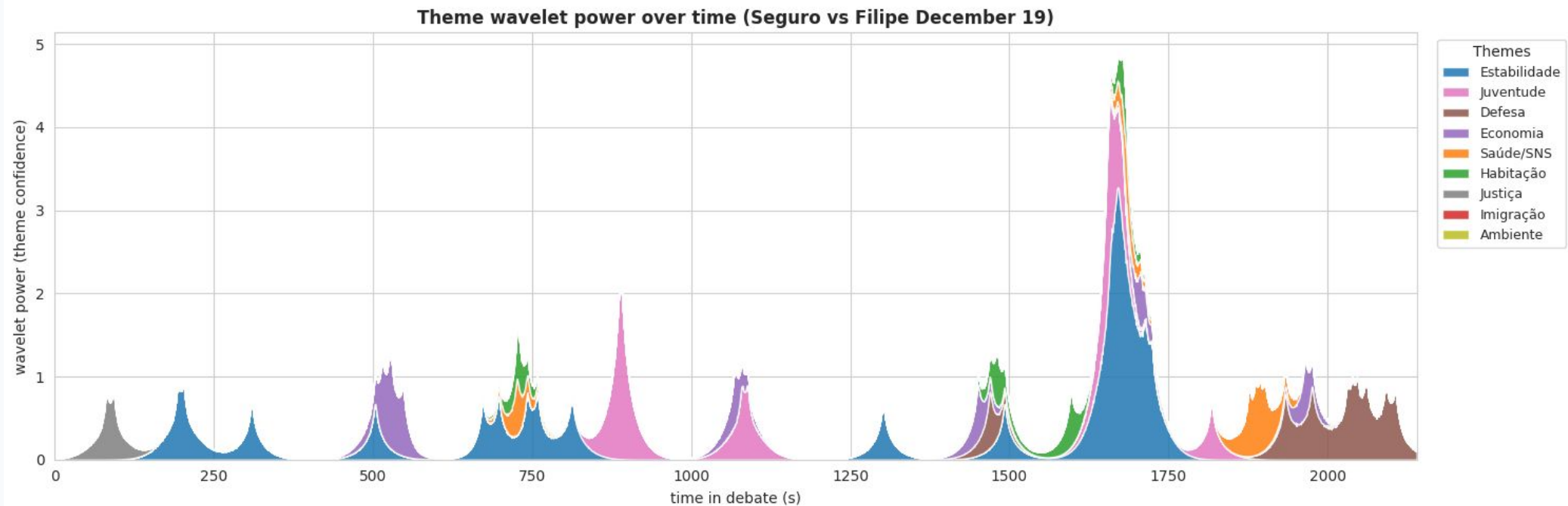
Debates vs. Newscasts Theme Comparison



Poll Correlation Scatterplots



- ## Wavelet Theme Power Distribution for one debate

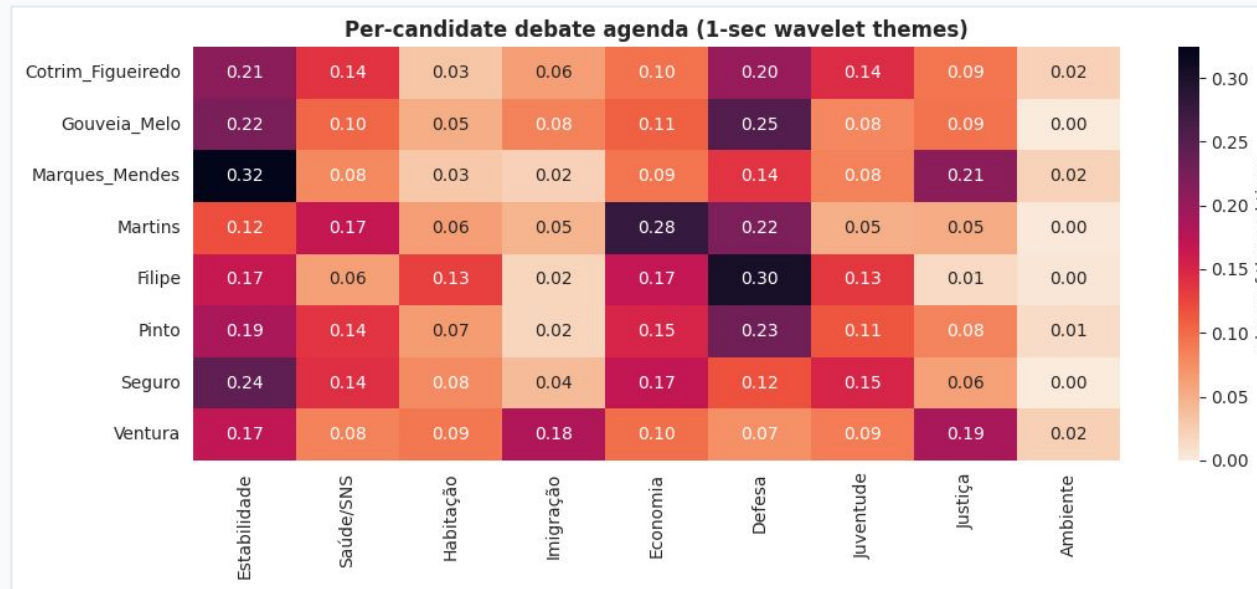


Campaign Themes

Cross-Campaign Theme Detection

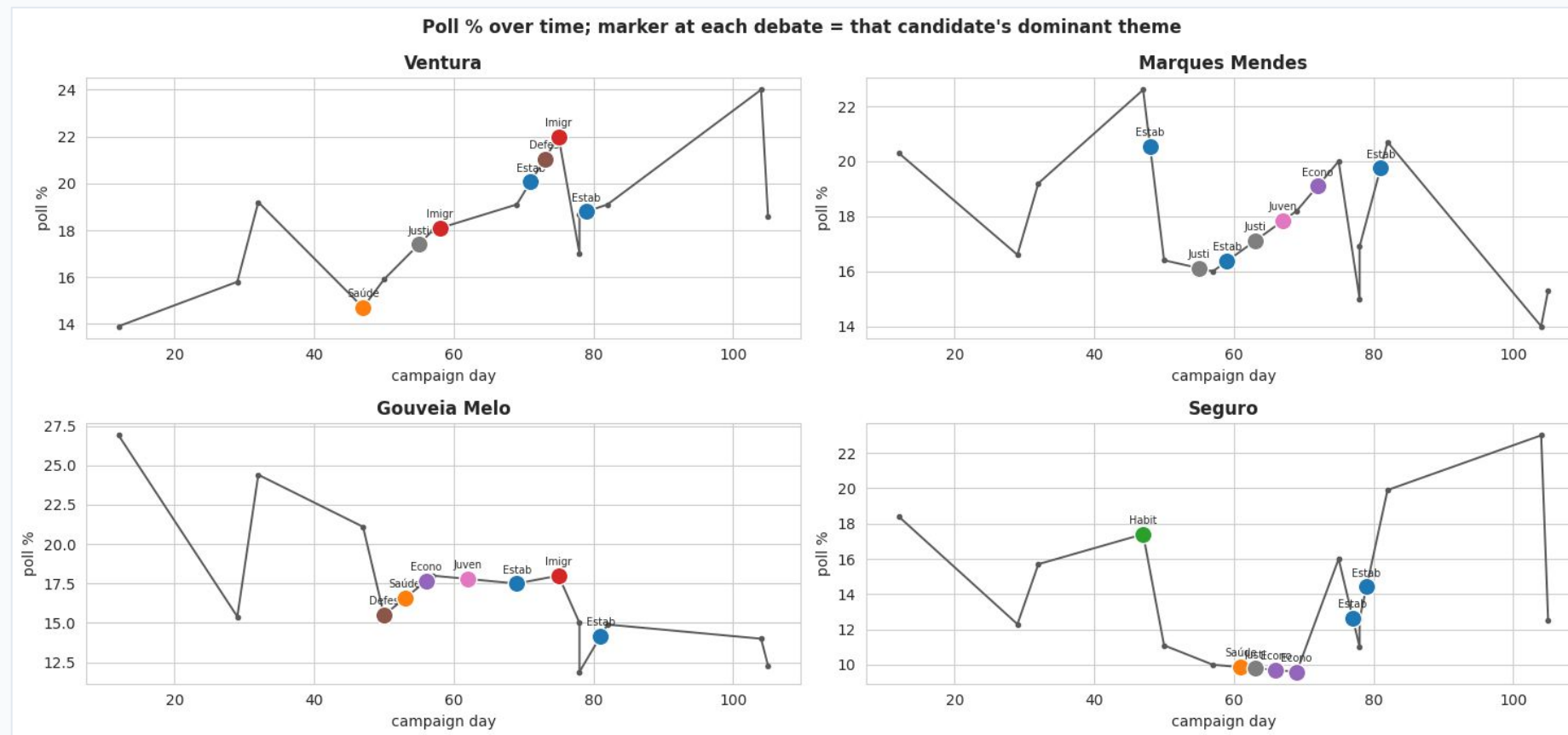
- Wavelet based 1-sec step keyword theme detector scored over all 28 debate transcripts
- Ambiente is notably absent;

Theme matrix per candidate

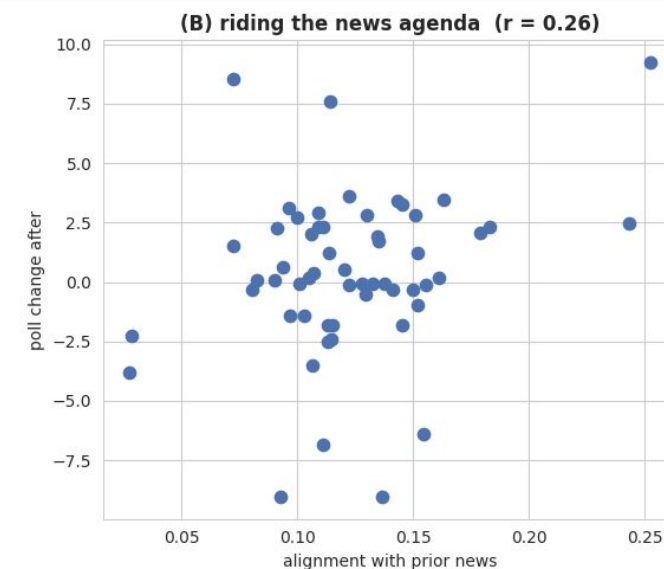
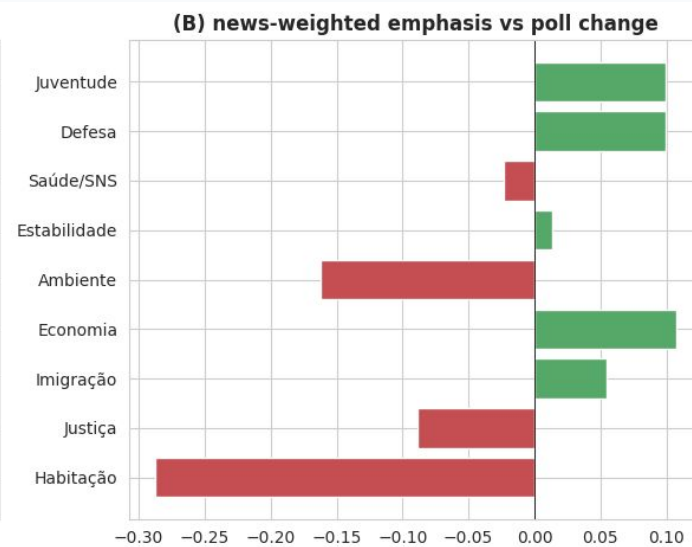
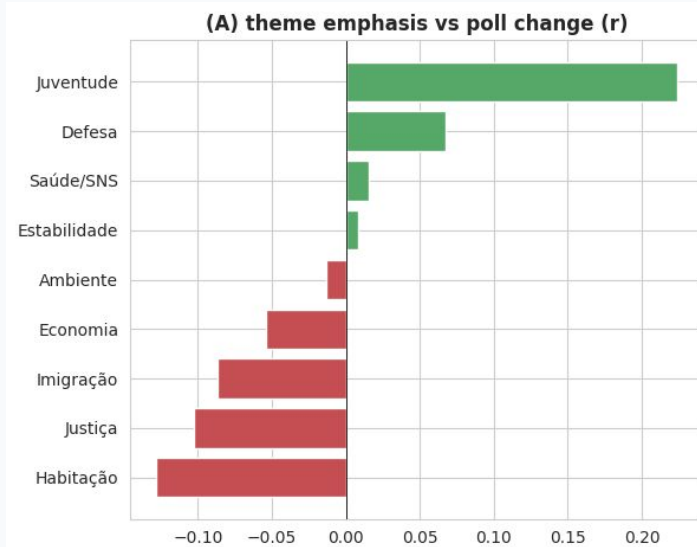


Campaign Themes over time

4 Candidates Main Theme Progression



Campaign Themes over time



Riding the news agenda is slightly poll-positive ($r \approx 0.25$) and **Juventude** is the main theme in doing it.

AI Attribution & Acknowledgments



Coding Assistant

Anthropic Claude Opus 4.7/4.8 Max
served as a coding assistant in the project.



Direction

The project's direction remained entirely human.



Velocity

AI assistance was leveraged to accelerate technical implementation and refactor messy logic.

Project Conclusions

Key Findings & Technical Highlights

- **Complementary Modalities:** Full agreement is only achieved through data fusion; no single modality is sufficient.

Media & Discourse

The press mirrors and tracks the election dynamics the best.

Limits

Regarding predictive power:

- 13 multi-agency polls: Themes do **not** predict polls.
- Hypothesis generating, not a predictor :(



Thank you !